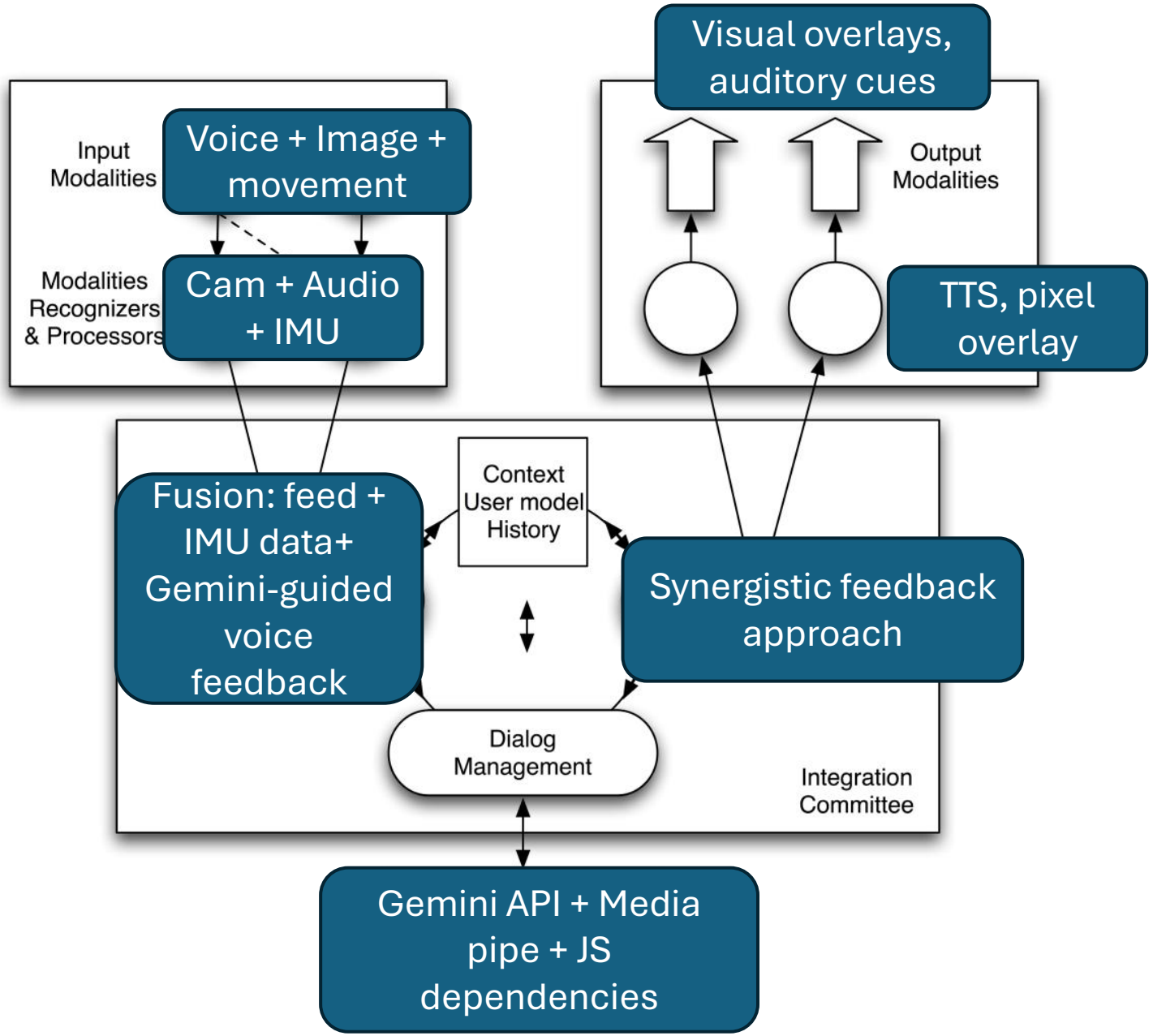
Impulse: Synergistic multimodal sports training

The problem:

- **Static training** (lack of gamification leads to low engagement)
- **Injury risk** (absence of RT, physio-informed posture correction)
- **Data saturation** (overwhelming raw metrics that act as noise more than actionable feedback)
- **Feedback void** (Inaccurate proprioceptive feedback and lack of post-execution quality feedback)



Architecture & phased implementation plan (14 weeks)

Input modalities fusion (Python): Optical recognition (video) + Hardware kinematics (IMU sensors) into a unified model.
Dialog management & context: Fusion of input (Python + HTML/JS), the dialog management oversees the Gemini API as an LLM-driven dialog manager to add dynamism to the feedback.

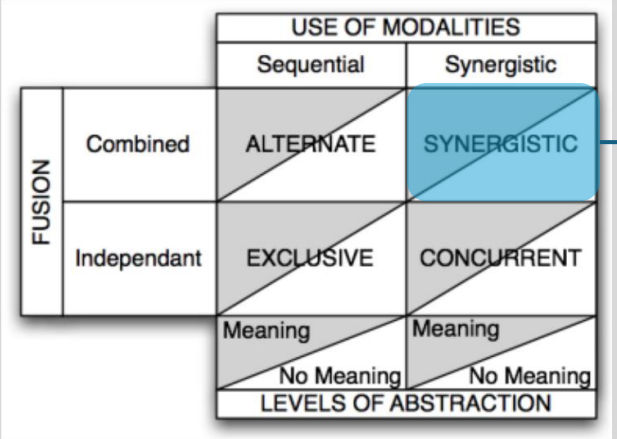
Output modalities fission (Python): Translating LLM's payload and mediapipe's inferences into a gamified visual and metrics with clear acoustic coaching cues.

We are combining multiple sensory information into an LLM-driven manager that provides insightful feedback.

CASE/CARE Models

CASE – "*how*" modalities are combined

CARE – "why" modalities are combined



Nigay and Coutaz 1993

Sprint 1 (visual Foundation) W4-W6
- Python integration of camera feed, audio processing and basic CV recognition.

Sprint 2 (Cognitive engine) W7-W9
- Integration of the Gemini API and GUI essentials

Sprint 3 (fusión + HW) W10-W14
- Wearable integration of IMU; polish of LLM inferences; potential integration of other systems*

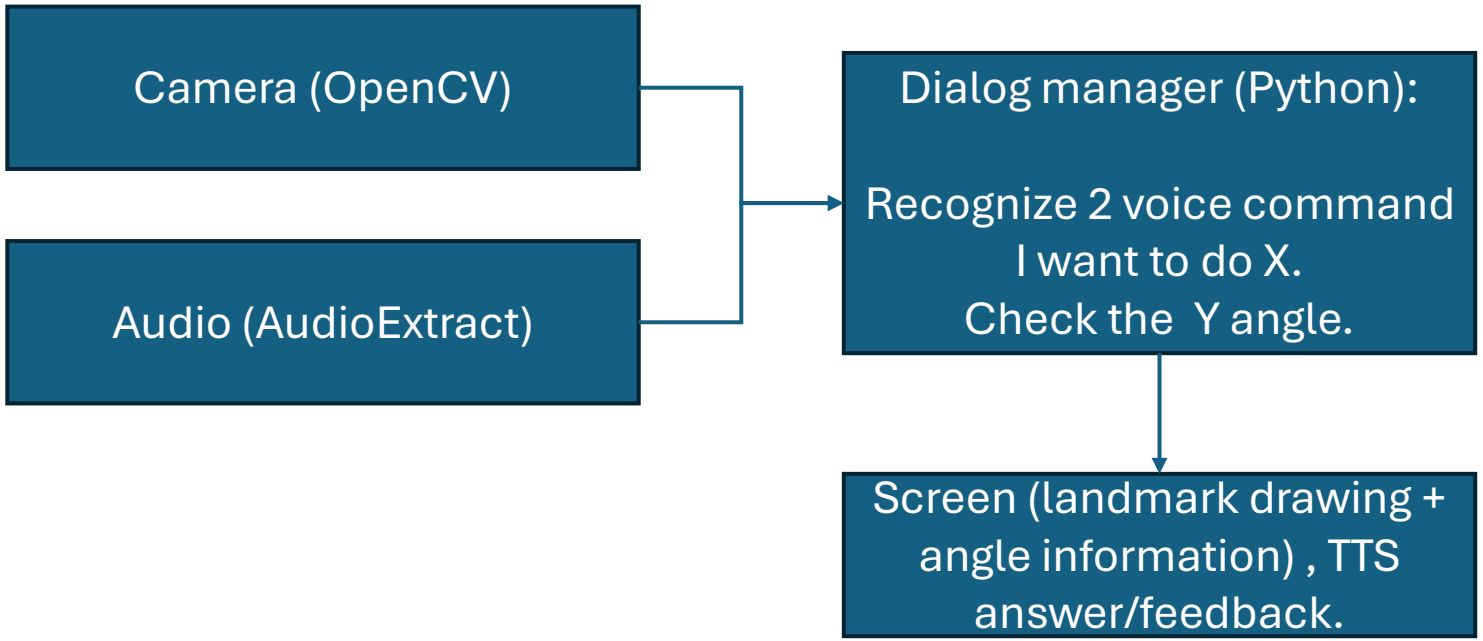
*potential systems to add: V-JEPA for physics informed inferences, Robust detections of IMU outside of angular velocity w. PINNs, Integration in a Flutter-based web App

Sprint 1: (visual Foundation) W4-W6

Responsibility	Assignee
Visual integration (OpenCV)	Mathilde, Sebastien
Audio integration (audio, PyGame?)	Simon, Santiago
Basic ML (Media Pipe)	Mathilde, Sebastien
LLM preparation (Gemini API)	Simon, Santiago
Architecture pipeline	Santiago, Simon

Target exercises: Bench press (mostly upper limbs, one cam), squats (two cams), dead lift (side view only).

Sprint 1 version



SQUATS (side view)

Back position (straight back)
Hips angle (hip flexion)
Depth of squat (thigh lower that parallel with ground)
NOT INWARD KNEES

Dead lift (sideview)

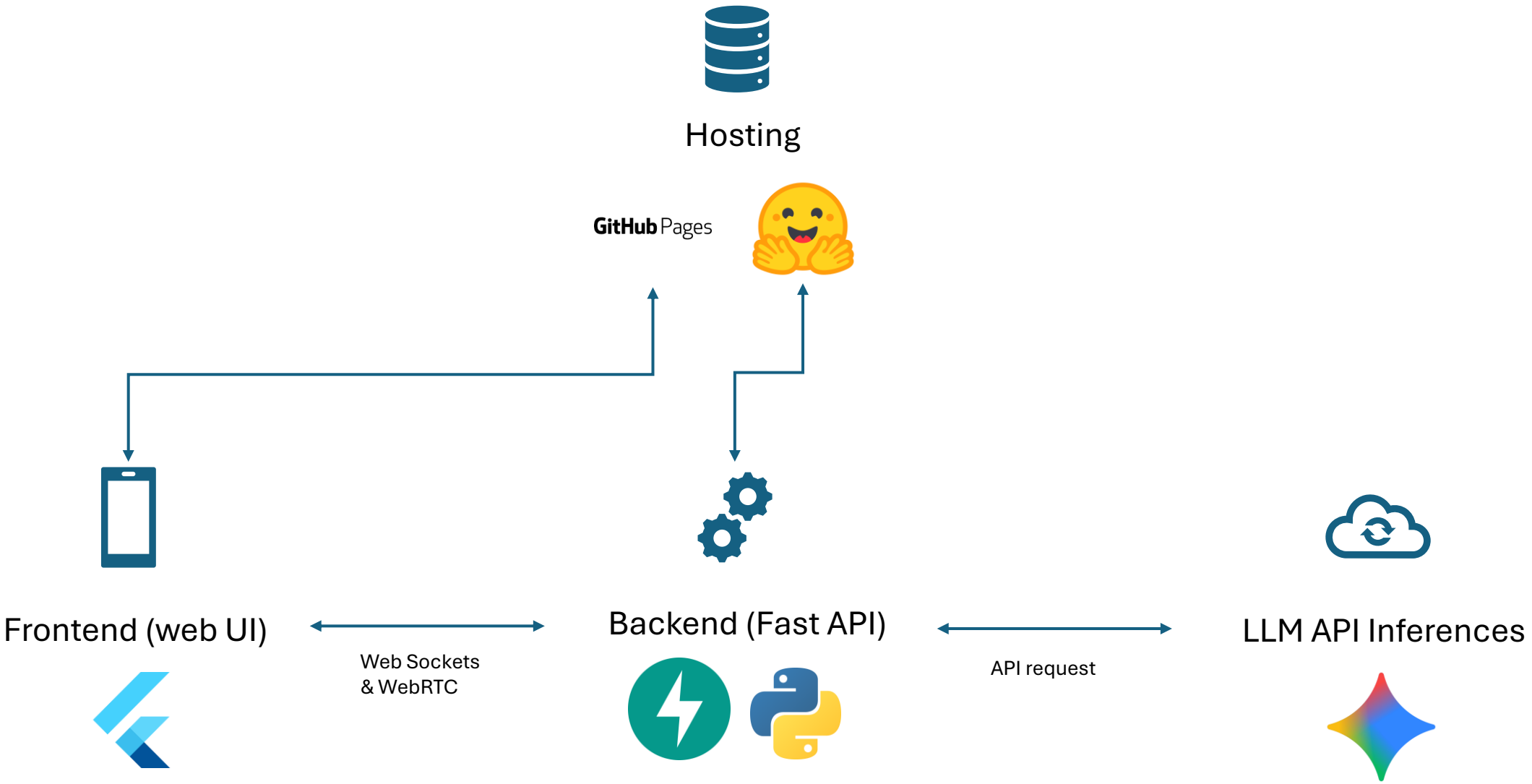
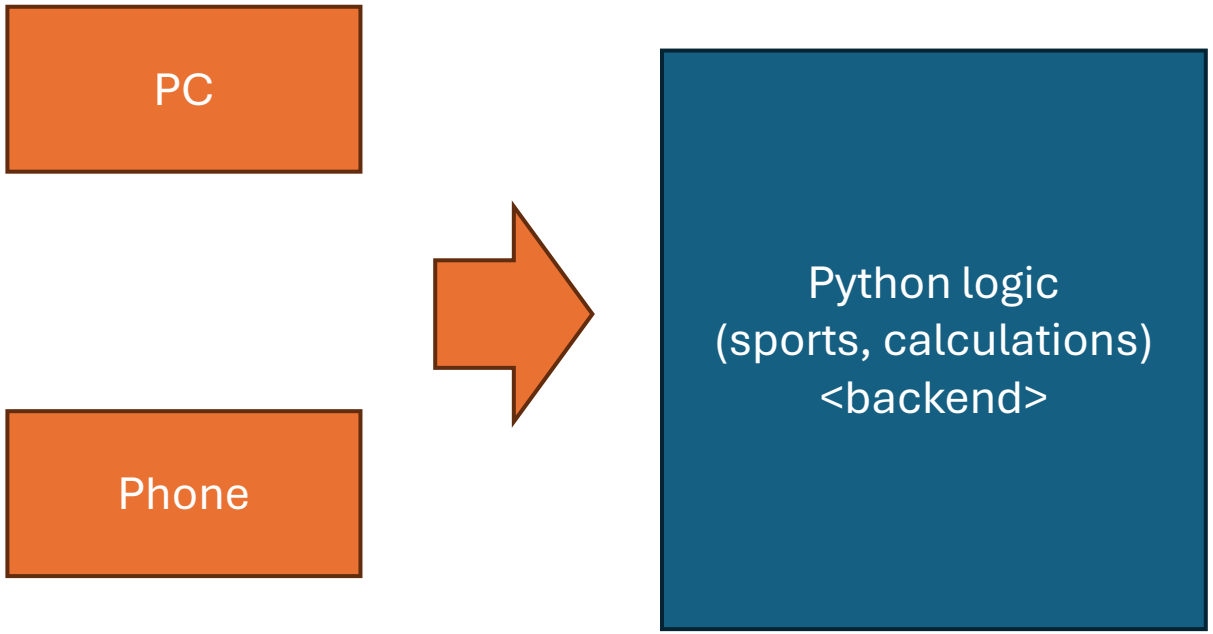
Back position (straight back)
Hip angle (hip flexion)
Knee flexion

Diagnostics (follow-up idea once all sprint 1 & 2 is done)

X measurements

1. Holding planck (s): core strength
2. Hold to a bar (s): Grip strength
3. Max O2 saturation level

4. Strength of lumbars vs abs (shirado-ito)


Hosting
GitHub Pages
Frontend (web UI)
Web Sockets
& WebRTC
Backend (Fast API)
API request
LLM API Inferences

Inference